

Chapter 13.8.

Critical point $f'(x) = 0$.

$f''(x) > 0 \rightarrow$ min value of func
 $f''(x) < 0 \rightarrow$ max value of func
 point of inflection $f''(x) = 0$.

Concave up $f''(x) > 0$.

Concave down $f''(x) < 0$

Q1



continuous

2nd
 order respect of derivative
 value same $f_{xy} = f_{yx}$

Hessian matrix.

$f(x, y) \rightarrow f_{xx}, f_{xy}, f_{yx}, f_{yy}$.

$$H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$$

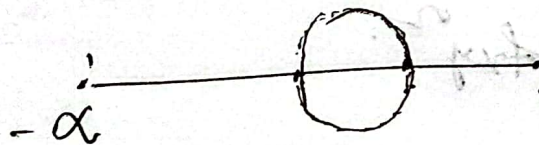
$$|H| = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}$$

For double variable

$f(x, y)$, we will get critical point at $f_x = 0$ and $f_y = 0$
critical point gives up maximum and minimum point & (concave and down) &

relative maxima (concave up)
absolute maxima max (check)

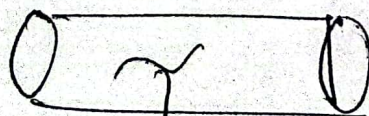
extrema (maxima + minima) $\Rightarrow$ $\frac{\partial f}{\partial x} = 0$ & $\frac{\partial f}{\partial y} = 0$



relative $\equiv$ inside the disk

absolute $\equiv$ all points in the domain of f .

Extreme value theorem:



both have both absolute (maxima and minima).

□ If a critical point neither maxima nor minima, then the point is saddle point.

$$D = f_{xx}f_{yy} - f_{xy}^2$$

$$H = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}$$

$$H = f_{xx}f_{yy} - f_{xy}f_{yx}$$

closed bounded, so $f_{xy} = f_{yx}$

$$H = f_{xx}f_{yy} - f_{xy}^2$$

(i) $D > 0$, $f_{xx} > 0 \rightarrow$ minimum

$D > 0$, $f_{xx} < 0 \rightarrow$ maximum

$D < 0$, saddle point.

$D = 0$, no operation/conclusion.

Example 3.

$$f(x, y) = 3x^2 - 2xy + y^2 - 8y.$$

$$f_x(x, y) = 6x - 2y.$$

$$f_y(x, y) = -2x + 2y - 8.$$

$$6x - 2y = 0.$$

$$-2x + 2y = 8.$$

$$4x = 8.$$

$$\boxed{x = 2}$$

$$-4 + 2y = 8.$$

$$2y = 12.$$

$$\boxed{y = 6}$$

critical point (2, 6).

$$f_{xx}(x, y) = 6.$$

$$f_{yy}(x, y) = 2.$$

$$f_{xy} = \frac{d}{dy} \left(\frac{dy}{dx} \right) = -2.$$

$$D = 6 \times 2 - (-2)^2 = 12 - 4 = 8 > 0 \quad \text{relative minima.}$$

$$f(x,y) = 4xy - x^4 - y^4.$$

$$\rightarrow f_x(x,y) = 4y - 4x^3$$

$$\rightarrow f_y(x,y) = 4x - 4y^3$$

$$4y - 4x^3 = 0.$$

$$4x - 4y^3 = 0$$

$$4x - 4y^3 = 0$$

$$4y - 4x^3 = 0$$

$$4(x-y) + 4(x^3-y^3) = 0.$$

$$4(x-y) + 4(x-y)(x^2+xy+y^2) = 0$$

$$\Rightarrow (x-y)(4 + 4x^2 + 4xy + 4y^2) = 0$$

$$\Rightarrow (x-y)$$

$$(x-y) = 0, 1, -1$$

$$(0,0), (1,1), (-1,-1)$$

$$D = f_{xx} - f_{yy} - f_{xy}^2$$

$$= -12x^2 - 12y^2 - 0$$

$$D =$$

$$f(x, y) = 3xy - 6x - 3y + 7.$$

$$f_x(x, y) = 3y - 6$$

$$f_y(x, y) = 3x - 3$$

$$3x - 3 = 0$$

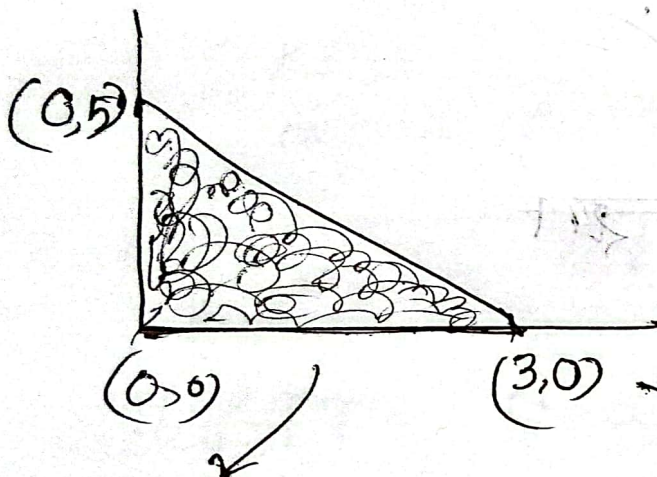
$$x = 1$$

$$3y - 6 = 0$$

$$y = 2$$

$$(1, 2)$$

$$(1, 2)$$



non-regional

एक भाग
जगह नहीं

$$f(x, 0) = -6x + 7$$

$$y = 0$$

$$f_x(x, 0) = -6$$

$$f_x(x, 0) = -6$$

$$= 6x + 7 = 0$$

$$x = -\frac{7}{6}$$

$$\left(-\frac{7}{6}, 0\right) \checkmark$$

$$x = 0$$

$$f_y(0, y) = -3y + 7$$

$$f_y(0, y) = -3$$

$$y = \frac{7}{3}$$

$$\left(0, \frac{7}{3}\right) \checkmark$$

$$x = -\frac{3}{5}(y-5)$$

$f_3(x)$

$$y = -\frac{5}{3}x + 5$$

~~$f_3(x) = 0$~~ → critical point

$$f(x, -\frac{5}{3}x + 5)$$

$$\equiv 3x \left(-\frac{5}{3}x + 5 \right) - 6x - 3 \left(-\frac{5}{3}x + 5 \right) + 7$$

$$= -5x^2 + 14x - 8$$

~~$(1, 2)$ बिन्दु पर~~

